

DESIGN OPTIMIZATION OF THE INSULATION DOMAIN IN A BUILDING WALL

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ABSTRACT

This work presents an innovative optimization approach for the thermal design of building walls, focusing on the geometry of the interface between structural and insulating materials. The proposed framework aims to reduce wall energy demand by optimizing this interface configuration. The physical problem is modelled by a two-dimensional transient heat transfer equation, solved using the DuFORT-FRANKEL finite difference scheme. The latter is an explicit scheme, unconditionally stable by construction, providing efficiency in capturing the unsteady thermal response of the wall. Boundary conditions include internal convection, external convection combined with shortwave radiation. These conditions vary both in time and space, enabling realistic simulations of daily climatic influences. The interface between materials is represented by a particle-based interpolation of control points, offering flexible geometric parametrizations. The objective function is defined as a ponderated function of the total heat flux across the internal surface of the wall, while the optimization, performed with a genetic algorithm, is carried out under the constraint of preserving the overall volume of both insulating and structural materials. Finally, case studies conducted for several Brazilian cities demonstrate the effectiveness of the proposed strategy and highlight the influence of climatic conditions on the optimized wall design.

KEYWORDS: Two-dimensional heat transfer, Numerical model, Building energy efficiency, Design problem, Topology optimisation

1. INTRODUCTION

Building envelopes account for a significant portion of energy consumption in the built environment [1]. Within this context, building energy efficiency represents a major concern, and building enclosures play a fundamental role in reducing thermal energy losses and, consequently, in minimizing the energy demand needed to maintain occupant comfort. However, conventional design approaches often rely on simplified assumptions: incident solar radiation and convective heat fluxes are assumed uniform across the façade surface (varying only in time, not in space), and wall configurations consist of planar layers with fixed interfaces between materials. These simplifications, though computationally tractable due to 1D modelling approaches, neglect important spatio-temporal heterogeneities that significantly influence wall thermal response, particularly in complex urban environments.

Recent studies have shown that spatial variations in boundary conditions cannot be overlooked when designing thermally efficient walls [2]. Wind regimes and shading effects, driven by the surrounding urban or natural environment, vary not only over time but also, and more importantly, across space. These non-uniform boundary conditions induce heterogeneous temperature fields within the wall, suggesting that they could be exploited to

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develop new insulation layouts rather than treated solely as constraints. This perspective becomes even more relevant in light of recent technological advances, such as the possibility of using 3D printing in construction [3]. Several works have begun to explore this direction, for instance by optimizing the thermal performance of lightweight concrete blocks [4] or by jointly enhancing the thermal and mechanical efficiency of building components [5]. In particular, [6] investigated a street-canyon configuration and optimized the insulation distribution by allocating materials over elementary subdomains, or [7] which optimizes façade geometry in a similar urban context by examining how street-canyon effects influence the optimization parameters [8]. However, all these contributions remain restricted to steady-state conditions, thereby neglecting the inherently transient nature of exterior climate loads in the optimization process.

This work proposes a novel framework for thermal wall design by optimizing the interface geometry between structural and insulating materials. The objective is to reduce wall energy demand through geometric parameter optimization, while preserving overall material volume. A parametric representations of the interface are introduced, based on particle interpolation using BÉZIER curves. Both enable flexible interface geometries governed by tractable sets of design parameters. The optimization problem is solved using a genetic algorithm NSGA-II [9], which offers substantial flexibility regarding the number of design parameters and enables an effective exploration of the search space to identify a global minimum.

The structure of the article is as follows. Section 2 presents the two-dimensional transient heat-transfer model for a wall exposed to an urban canyon environment, with particular attention given to the interface representation. Section 3 describes the design problem together with the adopted optimization strategy. Finally, Section 4 introduces a case study that illustrates the proposed approach for improving the energy efficiency of various building wall configurations under Brazilian climatic conditions.

2. PHYSICAL MODEL

2.1 Problem description

The physical problem involves two-dimensional transient heat transfer through a building wall defined by the domain $\Omega = [0, L] \times [0, H]$, where L [m] and H [m] represent the wall thickness and height, respectively. The boundary of the domain is denoted as $\Gamma = \bigcup_{j=1}^4 \Gamma_j$, where Γ_1 corresponds to the external surface, Γ_3 to the internal surface, and Γ_2, Γ_4 represent the top and bottom boundaries.

As illustrated in Fig. 1, the wall is located within an urban street canyon environment. The studied façade faces south and is positioned opposite to a building at distance ΔX [m] with height ΔY [m], which generates a time-dependent shadow on the studied wall. The façade consists of two materials separated by an interface Γ_i , defined implicitly by:

$$\Gamma_i(\mathbf{p}) = \left\{ \mathbf{x} \in \Omega \mid x = f_{\mathcal{B}_x}(s, \mathbf{p}), y = f_{\mathcal{B}_y}(s, \mathbf{p}), \mathbf{p} \in \Omega_p, s \in [0, 1] \right\} \quad (1)$$

where $f_{\mathcal{B}_x}$ and $f_{\mathcal{B}_y}$ are functions depending on the abscissa curvilinear s and governed by N_p design parameters $\mathbf{p} = (\mathbf{p}_1, \dots, \mathbf{p}_{N_p}) \in \Omega_p$.

The temperature field $T(\mathbf{x}, t)$ [K] within the wall is governed by the transient heat conduction equation:

$$\frac{\partial}{\partial x} \left(\lambda(\mathbf{x}) \frac{\partial T}{\partial x} \right) + \frac{\partial}{\partial y} \left(\lambda(\mathbf{x}) \frac{\partial T}{\partial y} \right) = c(\mathbf{x}) \frac{\partial T}{\partial t}, \quad \forall \mathbf{x} \in \Omega, \quad \forall t > 0 \quad (2)$$

where λ [W.m⁻¹.K⁻¹] is the heat conductivity and c [J.m⁻³.K⁻¹] the volumetric heat capacity. At the initial time, the wall is assumed to be in a state consistent with the prescribed boundary conditions:

$$T = T_0(\mathbf{x}), \quad \forall \mathbf{x} \in \Omega, \quad t = 0 \quad (3)$$

The thermal interaction at the external surface Γ_1 combines conductive, convective, and radiative transfer. It

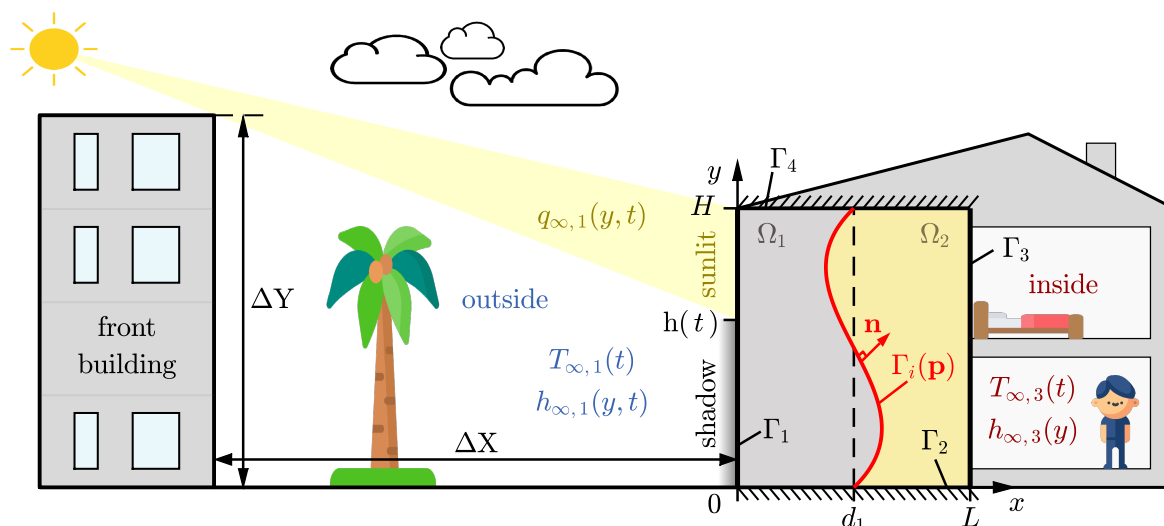


Fig. 1 *Illustration of the urban street canyon configuration.*

can be expressed as:

$$-\lambda(\mathbf{x}) \frac{\partial T}{\partial \mathbf{x}} = h_{\infty,1}(y, t) (T_{\infty,1}(t) - T) + q_{\infty,1}(y, t), \quad \forall \mathbf{x} \in \Gamma_1 \quad (4)$$

where $T_{\infty,1}$ [K] is the external ambient temperature, $h_{\infty,1}$ [W.m⁻².K⁻¹] is the convective surface coefficient, and $q_{\infty,1}$ [W.m⁻²] is the incident short-wave radiation flux decomposed as follows:

$$q_{\infty,1}(y,t) = \alpha \left(q^{\text{dr}}(y,t) + q^{\text{df}}(t) + q^{\text{rf}}(t) \right),$$

where q^{dr} [W.m⁻²], q^{df} [W.m⁻²] and q^{rf} [W.m⁻²] are the direct, diffuse and reflected components. α [-] is the absorptivity of the wall and is set to 0.8. Long-wave radiative flux is not taken into account here. The convective heat transfer coefficient varies with height and wind velocity v_{∞} [m.s⁻¹] according to empirical correlation of [10]:

$$h_{\infty,1}(y, t) = h_{10} + h_{11} \cdot \frac{v_{\infty}(t)}{v_0} \cdot \left(\frac{y}{y_0} \right)^{\beta} \quad (5)$$

where the surface coefficients are set to $h_{10} = 5.82 \text{ W.m}^{-2}.\text{K}^{-1}$ and $h_{11} = 3.96 \text{ W.m}^{-2}.\text{K}^{-1}$, the velocity variation coefficient to $\beta = 0.32$, and the reference quantities to $v_0 = 1 \text{ m.s}^{-1}$ and $y_0 = 65.33 \text{ m}$. The incident radiation flux varies in space and time due to the shading effects of the opposite building. The position of the sun and the orientation of the facade give the instantaneous height of the shadow h , which is therefore obtained from the fraction of illuminated surface area calculated using the pixel counting technique [11] used in the Domus software. This results in a total flux $q_{\infty,1}$ including the diffuse, reflected, and direct components according to height, as detailed in the work of [2]. Finally, the external ambient temperature, wind velocity, and radiative flux components are directly extracted from the meteorological data.

On the internal surface, the thermal balance is achieved solely through conductive and convective flux:

$$\lambda(\mathbf{x}) \frac{\partial T}{\partial \mathbf{x}} = h_{\infty,3}(y) \cdot (T_{\infty,3}(t) - T), \quad \forall \mathbf{x} \in \Gamma_3 \quad (6)$$

where the interior convective coefficient varies with height [12]:

$$h_{\infty,3}(y) = h_{30} + (h_{31} - h_{30}) \cdot \sin\left(\frac{\pi y}{H}\right) \quad (7)$$

with the surface coefficients $h_{30} = 0.1 \text{ W.m}^{-2}.\text{K}^{-1}$ and $h_{31} = 3 \text{ W.m}^{-2}.\text{K}^{-1}$. It should be noted that the

long-wavelength radiative component was not included in the present model to avoid introducing additional complexity. The internal temperature $T_{\infty,3}$ represents a conditioned environment with heating and cooling systems:

$$T_{\infty,3}(t) = \begin{cases} T_{\text{set}}^w, & \text{si } T_3(t) \leq T_{\text{set}}^w, \\ T_3(t), & \text{si } T_{\text{set}}^w < T_3(t) < T_{\text{set}}^s, \\ T_{\text{set}}^s, & \text{si } T_3(t) \geq T_{\text{set}}^s \end{cases} \quad T_3(t) = T_{\text{set}}^w + \frac{T_{\infty,1}(t) - T_{\text{heat}}}{T_{\text{cool}} - T_{\text{heat}}} (T_{\text{set}}^s - T_{\text{set}}^w) \quad (8)$$

where $T_{\text{set}}^w = 19^\circ\text{C}$ and $T_{\text{set}}^s = 24^\circ\text{C}$ are heating and cooling setpoints, and $T_{\text{heat}} = 17^\circ\text{C}$ and $T_{\text{cool}} = 27^\circ\text{C}$ are external activation thresholds.

At the top Γ_4 and bottom Γ_2 of the wall, perfect insulation is assumed, which is equivalent to imposing adiabatic boundary conditions:

$$-\lambda(\mathbf{x}) \frac{\partial T}{\partial y} = 0 \quad \forall \mathbf{x} \in \Gamma_2, \quad \lambda(\mathbf{x}) \frac{\partial T}{\partial y} = 0 \quad \forall \mathbf{x} \in \Gamma_4 \quad (9)$$

Finally, at the material interface Γ_i , both temperature and heat flux continuity must be enforced, which can be written as:

$$T(\mathbf{x} - \varepsilon) = T(\mathbf{x} + \varepsilon), \quad \forall \mathbf{x} \in \Gamma_i \quad (10a)$$

$$-\lambda(\mathbf{x}) \frac{\partial T}{\partial \mathbf{n}} \Big|_{\mathbf{x} - \varepsilon} = \lambda(\mathbf{x}) \frac{\partial T}{\partial \mathbf{n}} \Big|_{\mathbf{x} + \varepsilon}, \quad \forall \mathbf{x} \in \Gamma_i \quad (10b)$$

where $\varepsilon \rightarrow 0$, $\mathbf{n}$ is the interface normal from Ω_2 to Ω_1 , and $\mathbf{x} \pm \varepsilon$ denote positions on either side of the interface.

To solve the heat-transfer problem, we use the numerical model introduced in [13], which combines the DuFort-Frankel scheme with the Ghost Fluid Method [14] to handle the interface problem efficiently.

2.2 Thermophysical properties and interface definition

The interface Γ_i separating the two material domains can be described using a flexible particle-based representation. In this approach, the interface is defined through interpolation between $N_\bullet$ control points with coordinates $\mathbf{p}_i = (x_i, y_i)$. Among the available interpolation methods, we employ BÉZIER parametric curves. BÉZIER curve position is expressed separately for the horizontal and vertical coordinates using BERNSTEIN polynomials $\mathcal{B}$ and according to the curvilinear abscissa s , such that:

$$f_{\mathcal{B}_x}(s, \mathbf{p}) = \sum_{i=1}^{N_\bullet} \mathcal{B}_{i-1}^{N_\bullet}(s) \cdot x_i + A(\mathbf{p}), \quad (11a)$$

$$f_{\mathcal{B}_y}(s, \mathbf{p}) = \sum_{i=1}^{N_\bullet} \mathcal{B}_{i-1}^{N_\bullet}(s) \cdot y_i, \quad \forall s \in [0, 1] \quad (11b)$$

Here, $A(\mathbf{p})$ is the volume constraint term, imposing the volumes of the two layers Ω_1 and Ω_2 to remain equal to those corresponding to a flat interface at $x = d_1$. After expansion, it takes the form:

$$A(\mathbf{p}) = d_1 - \frac{N_\bullet}{H} \sum_{j=1}^{N_\bullet} x_j \cdot \left[\sum_{i=1}^{N_\bullet-1} (y_i - y_{i-1}) \cdot M_{ji} \right], \quad \text{with} \quad M_{ji} = \int_0^1 \mathcal{B}_{j-1}^{N_\bullet}(s) \cdot \mathcal{B}_{i-1}^{N_\bullet-1}(s) ds$$

If we define $\mathbf{x}_{\Gamma_i} = (x_{\Gamma_i}, y_{\Gamma_i})$ as a point in Ω belonging to the interface Γ_i , we denote by $s^\star$ the corresponding curvilinear abscissa. Thus, for a point $(\mathbf{x}_\Gamma, s^\star)$ at the interface Γ_i we can introduce the distance:

$$\delta = x - f_{\mathcal{B}_x}(s^\star, \mathbf{p}) \quad (12)$$

In other words, δ represents the horizontal distance between the point $\mathbf{x}_{\Gamma_i}$ and a point $\mathbf{x} = (x, y)$ in the domain

Ω . Before the interface this distance is negative, and after the interface it is positive.

Then, each material layer is characterized by constant thermal properties. Denoting by (λ_1, c_1) the properties in the first layer Ω_1 and by (λ_2, c_2) those in the second layer Ω_2 , the spatial distribution of these properties throughout the wall domain is defined using the Heaviside function $\mathcal{H}$:

$$\lambda(\mathbf{x}) = \lambda_1 \cdot (1 - \mathcal{H}[\delta]) + \lambda_2 \cdot \mathcal{H}[\delta], \quad \forall \mathbf{x} \in \Omega \quad (13a)$$

$$c(\mathbf{x}) = c_1 \cdot (1 - \mathcal{H}[\delta]) + c_2 \cdot \mathcal{H}[\delta], \quad \forall \mathbf{x} \in \Omega \quad (13b)$$

Figure 2 illustrates examples of interface shapes that can be produced using BÉZIER curves with three and four particles, respectively. The wall under consideration has dimensions $L = 15\text{cm}$ and $H = 3\text{m}$. The interface is designed so that the volumes of the two layers correspond to those obtained for a flat interface located at $d_1 = 10\text{cm}$, under an internal wall insulation (IWI) configuration, as indicated by the dashed line in Fig. 2a and Fig. 2b.

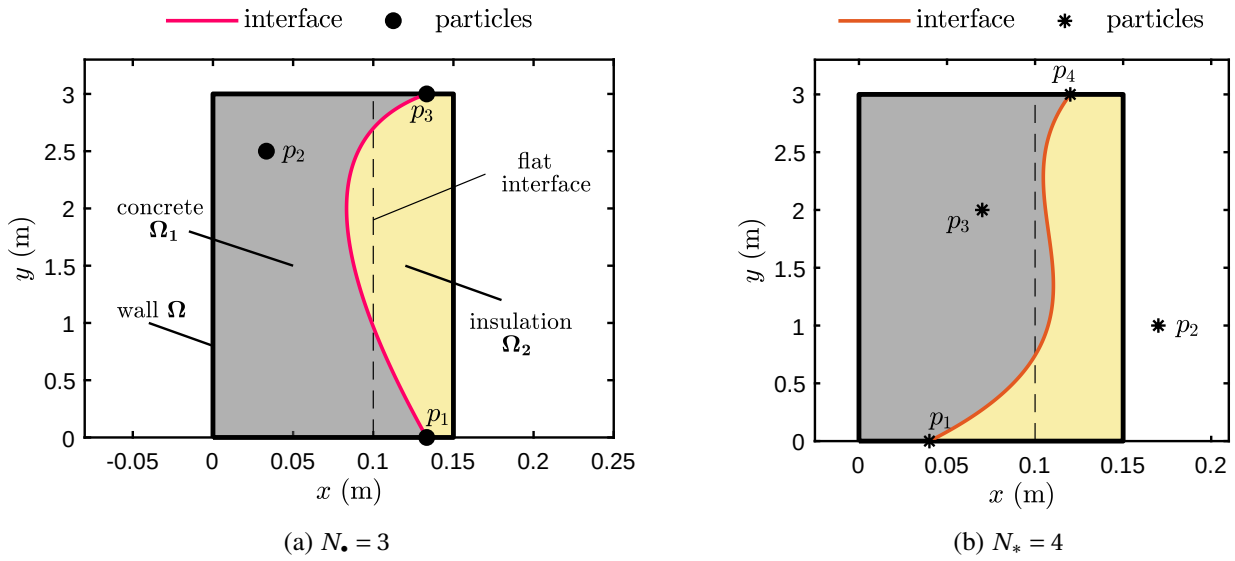


Fig. 2 Illustration of the interface Γ_i definition using BÉZIER approach for three (a) and four (b) particles, and their flat reference interface.

3. THERMAL DESIGN PROBLEM

3.1 Definition

The purpose is to optimize the energy efficiency of the facade by finding the optimal values of the parameters $\mathbf{p}^*$ that governed the shape of the interface between the materials:

$$\mathbf{p}^* = \arg \min_{\mathbf{p} \in \Omega_p} \mathcal{C}$$

The optimization criterion aims at reducing the time-integrated of the net consumed work w [W]. This value considers the wall as a thermal engine system, as described in [15]. Thus, the heat flux crossing the inside face of the wall is weighted by the ratio of external to internal temperatures, reflecting the potential for mechanical work according to the second law of thermodynamics. The resulting cost function $\mathcal{C}$ [J] is therefore defined as:

$$\mathcal{C} = \int_0^{t_f} w(t) dt, \quad \text{with} \quad w(t) = \left(\int_0^H -\lambda(\mathbf{x}) \cdot \frac{\partial T}{\partial x} dy \right) \cdot \left(\frac{T_{\infty,1}(t)}{T_{\infty,3}(t)} - 1 \right) \quad (14)$$

The problem is then posed as finding the interface geometry parameters $\mathbf{p}^*$ that minimize this cost function. The optimization search is subject to three main constraints that define the admissible domain for the parameters. First, the interface must remain entirely within the wall thickness while maintaining a minimum clearance δ_l [m] from the boundaries, ensuring adequate layer thickness for proper numerical discretization. Second, the equal-volume constraint enforces that the total amount of each material is preserved regardless of interface deformation. This is automatically satisfied by the interface definition with the term $A(\mathbf{p})$ eq. (11b). Third, the parameters themselves are bounded by user-specified lower and upper limits $\mathbf{p}_{\min}$ and $\mathbf{p}_{\max}$ to further restrict the search space based on physical and numerical requirements.

These three restrictions define jointly the admissible domain for the parameters: spatial confinement within prescribed margins, volume conservation, and parameter bounds are all simultaneously enforced during the optimization procedure.

3.2 Optimization strategy

The optimization is performed using the NSGA-II genetic algorithm. Convergence is monitored through two criteria: the relative variation of the objective function between successive generations $\delta\mathcal{C}$ and the spatial dispersion of design parameters within the population δd , calculated as the mean of the EUCLIDEAN norm of the $\mathbf{p}_i$ of each individual i with the mean $\mathbf{p}_c$ of the population.

$$\varepsilon_g = \max(\delta\mathcal{C}_g, \delta d_g) \quad \forall g \in \{2, \dots, N_g\}$$

with:

$$\delta\mathcal{C}_g = \frac{|\mathcal{C}_{g-1} - \mathcal{C}_g|}{\mathcal{C}_{g-1}}, \quad \delta d_g = \frac{2}{N_i} \sum_{i=1}^{N_i} \|\mathbf{p}_i - \mathbf{p}_c\|_2$$

The algorithm terminates when the criteria falls below $\varepsilon_g < \varepsilon = 10^{-3}$. According to the particles definition of the interface, the number of optimization variables does not strictly match the total number of particles describing the interface geometry. This is because the top and bottom anchor points are constrained to fixed vertical coordinates, namely $y_1 = 0$ and $y_{N_\bullet} = H$. Consequently, only the intermediate control points, together with the horizontal coordinates of the two attached points, are effectively optimized. The resulting number of design variables is therefore $N_p = 2(N_\bullet - 1)$. This requirement prevents the interface from having anchor points on the vertical faces of the domain, thereby ensuring the continuous presence of two distinct layers, which is necessary for maintaining the structural integrity of the wall.

4. CASE STUDY

4.1 Description

The case study considers a north facing building facade located in Rio de Janeiro, Brazil, within an urban canyon configuration, as illustrated in Fig. 1. The opposing building has a height of $\Delta Y = 3$ m and is positioned at a distance $\Delta X = 3$ m. The wall has a height of $H = 3$ m and a width of $L = 15$ cm. It consists of a concrete layer combined with glasswool insulation, whose corresponding properties are summarized in Table 1. Each layer is assigned a volume identical to that of the reference configuration, which corresponds to a flat interface located at d_1 .

Table 1 Thermal properties of the material composing the wall.

<i>Domain</i>	<i>Material</i>	<i>Thermal conductivity</i> [W.m ⁻¹ .K ⁻¹]	<i>Volumetric heat capacity</i> [MJ.m ⁻³ .K ⁻¹]	<i>Length in reference case</i> [cm]
Ω_1	concrete	2	1.94	10
Ω_2	insulation	0.04	0.026	5

All simulations conducted in this study cover a full year ($t_f = 365$ d) and employ a time step of $\Delta t = 30$ s. The

meteorological data, extracted from Meteonorm [16] and initially provided at 6min intervals, are resampled to this temporal resolution through linear interpolation. These conditions are illustrated in Fig. 3. Figure 3a reveals a non-uniform distribution of radiative flux along the façade height, characteristic of the urban-canyon configuration. In winter, the upper part of the wall receives a higher flux than the lower part. This occurs because the lower part is more strongly shaded, resulting in a reduced contribution of direct radiation when the sun is lower in the sky. Conversely, during this period, the upper portion experiences a higher flux, as the incident solar rays strike the façade with a smaller inclination angle compared to summer. Figure 3b further shows that the climatic conditions in Rio de Janeiro lead to cooling needs throughout the entire year. Regarding the external convective coefficient, Fig. 3c, no pronounced annual trend is observed. Instead, the variations occur mainly at the daily scale, with particularly strong fluctuations for the upper part of the façade.

The computational domain is discretized using $N_x = N_y = 100$ grid points in each spatial direction, which ensures convergence and good accuracy of the results.

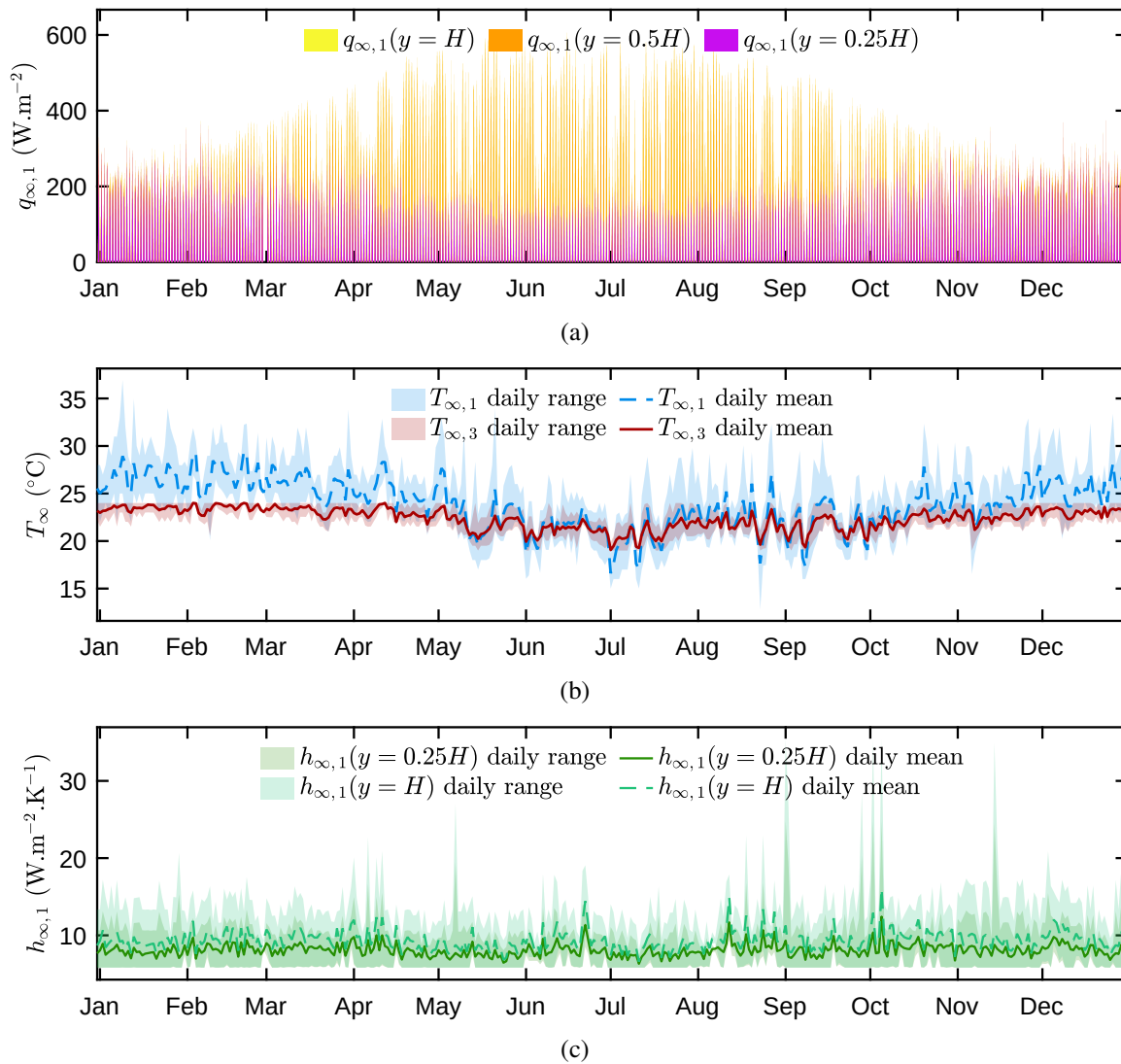


Fig. 3 Variations of external incident short-wave radiative flux on the facade (a), external and internal air temperatures (b), and external convective heat transfer coefficients (c) over time.

4.2 Wall insulation configuration

The optimization problem is first solved for two distinct insulation configurations: internal wall insulation (IWI) and external wall insulation (EWI). The algorithm operates on a population of $N_i = 42$ individuals evolving over a maximum of $N_g = 80$ generations, with crossover rate $c_r = 80\%$ and mutation rate $m_r = 20\%$.

Both simulations converged following the strategy described in Section 3.2, with the convergence history reported in Fig. 4. It is worth noting that the first criterion $\delta\mathcal{C}$ related to the cost function reaches its prescribed threshold much more rapidly than the second criterion δd , which concerns the dispersion of the control parameters. This behavior stems from the nature of BÉZIER curves, which can produce very similar interface shapes even when the control points are not located at exactly the same positions. Ultimately, the two optimization processes converged after 48 generations for the IWI case and 64 generations for the EWI case.

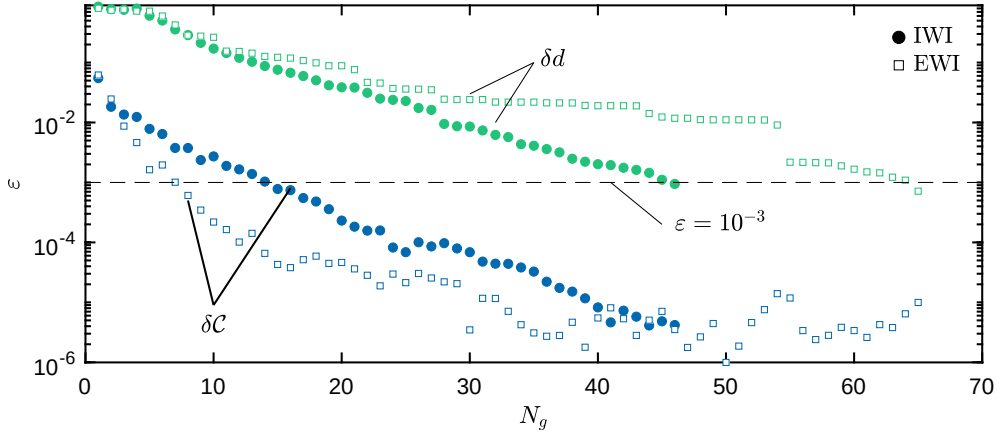


Fig. 4 Convergence of the optimization method for the two insulation configurations.

The resulting optimal interfaces are shown in Fig. 5a and Fig. 5c. For the IWI case, the optimal particle set is:

$$\mathbf{p}^*_{\text{IWI}} = (0.217, (0.186, 0.514), (0.119, 2.093), 0.217).$$

The corresponding interface exhibits a single inflection point located above mid-height, around $y = 1.7$ m. In contrast, the EWI configuration is characterized by the particle set:

$$\mathbf{p}^*_{\text{EWI}} = (0.094, (0.118, 0.992), (0.147, 1.956), 0.097),$$

which produces also an inflection point located around $y = 1.9$ m. In addition, Fig. 5b and Fig. 5d present the evolution of the average annual surface heat flux, on the external $\overline{q}_1$ and internal $\overline{q}_3$ faces of the wall respectively. These fluxes are compared with those obtained for the corresponding reference configurations. Focusing first on the reference IWI and EWI cases, the IWI configuration produces a much larger flux amplitude on the exterior surface than EWI, while the resulting magnitudes on the interior surface remain similar. This behavior is attributed to the higher thermal inertia of concrete compared to the insulation layer. Indeed, a portion of the energy absorbed during warm periods is stored within the concrete and subsequently released during cooler periods, towards the exterior in the IWI configuration, whereas in the EWI case the release occurs predominantly toward the interior. This effect is clearly illustrated by the negative mean flux observed over a portion of the façade in the IWI case Fig. 5b.

This behavior can also be observed in Fig. 6, where the concrete layer experiences a strong temperature rise under internal insulation, followed by a significant outward heat release during nighttime. The opposite pattern is visible for the EWI configuration. Finally, Fig. 6 highlights the vertical variability in temperature distribution along the façade, which explains how the optimal interface adapts to heat fluxes, particularly by placing more insulation in areas where radiative effects are strongest.

For both optimal interface configurations, a reduction of the surface flux is observed on both sides of the wall. This reduction enhances internal comfort, as the building is almost exclusively in cooling demand throughout the year. However, the optimal IWI configuration achieves a larger decrease in flux compared with its EWI counterpart. As a result, the annual energy efficiency improves by 5.70% for the optimal IWI configuration, compared with 1.69% for the optimal EWI configuration.

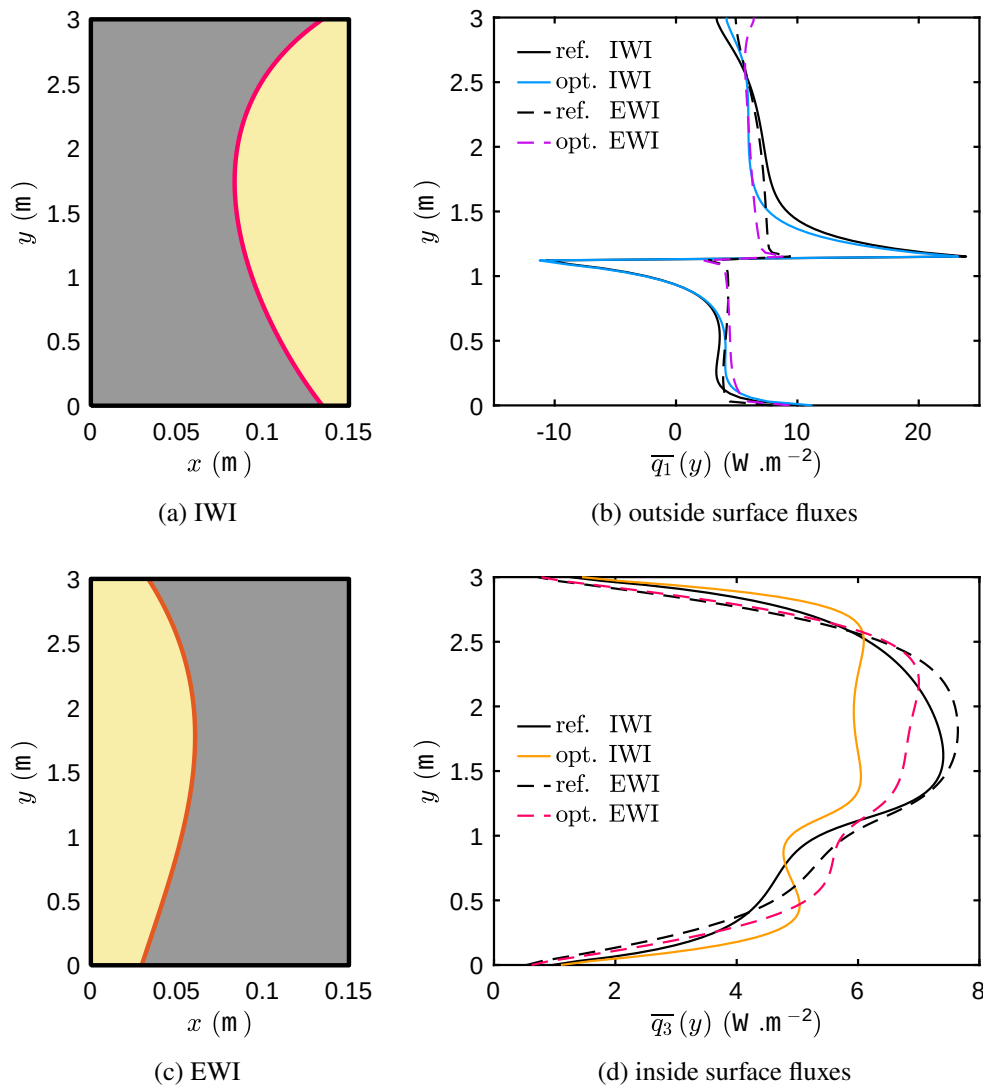


Fig. 5 Illustration of optimal interfaces in IWI (a) and EWI (c) configurations, and surface fluxes generated on the external (b) and internal (d) faces by these optimal configurations and their reference with flat interface.

4.3 Effect of façade orientation

The study is subsequently extended to six Brazilian cities to assess the influence of different climatic conditions. In addition, a west facing orientation of the façade is considered alongside the north facing configuration.

A visualization of the results is provided in Fig. 7 for the IWI configuration. First, Fig. 7a shows that improvement levels vary substantially for north facing walls. Cities such as Curitiba and São Paulo exhibit significant gains, 37% and 21%, respectively, whereas the other locations show only modest improvements, remaining around 5%. This can be explained by the fact that these two cities are located at much higher altitudes than the others, resulting in cooler ambient temperatures. As a consequence, solar radiation has a stronger relative influence on heat transfer through the wall. The optimal interface shapes for north facing wall shown in Fig. 7b are consistent with these observations. Moreover, Manaus, Teresina, and Rio de Janeiro exhibit similar interface configurations, with a larger amount of insulation concentrated around mid-height. These cities have very hot climates with substantial solar exposure over the upper two thirds of the façade, and the interface shape acts to mitigate these heat gains. In contrast, Curitiba and São Paulo show very limited insulation in the upper region of the wall, suggesting that the interface favours maximizing solar gains. Florianópolis also displays this trend, though less markedly, likely due to its intermediate climate, warmer than São Paulo but receiving somewhat less solar irradiance. For the west facing orientation, the optimal interfaces shown in Fig. 7d exhibit shapes very similar to those obtained for the north facing façade. All cities display improvements of comparable

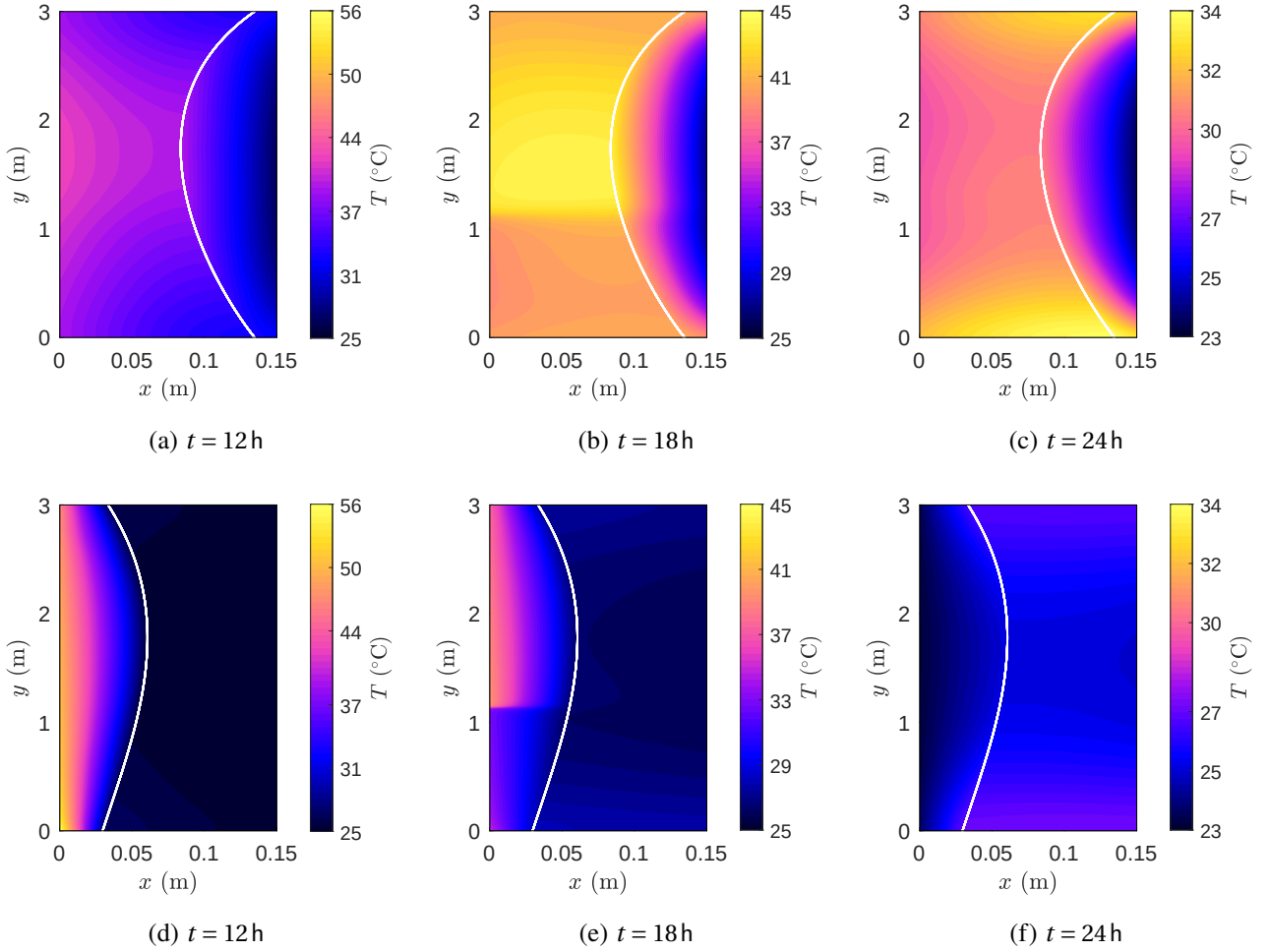


Fig. 6 Temperature field on December 1st at several times of the day for the optimal IWI (a-c) and EWI (d-f) configurations.

magnitude to the north orientation, except for Curitiba and São Paulo, where the gains decrease to 15% and 7%, respectively. This indicates that interface shape optimization can remain effective across multiple façade orientations.

The improvement values are also reported in detail in Table 2, together with the results obtained for the EWI configuration. As observed for the IWI configuration, the magnitude of improvement for the EWI configuration remains comparable between the two orientations, except for Curitiba, which exhibits a noticeably higher improvement for the north facing façade than for the west facing one. Focusing on the interface shapes shown in Fig. 8, one can see that Manaus, Teresina, and Rio de Janeiro again produce similar profiles, which appear as symmetric counterparts of the IWI cases. In contrast, São Paulo and Curitiba now allocate more insulation toward the upper part of the wall, in opposition to the IWI results, and this trend is consistent across both orientations. Florianópolis exhibits an interface shape that closely resembles São Paulo for the north facing façade, but aligns more closely with Rio de Janeiro for the west facing façade, thus revealing two completely distinct optimization strategies.

Overall, the IWI configuration delivers larger performance gains. Nevertheless, for both IWI and EWI, the two orientations remain relevant for interface optimization.

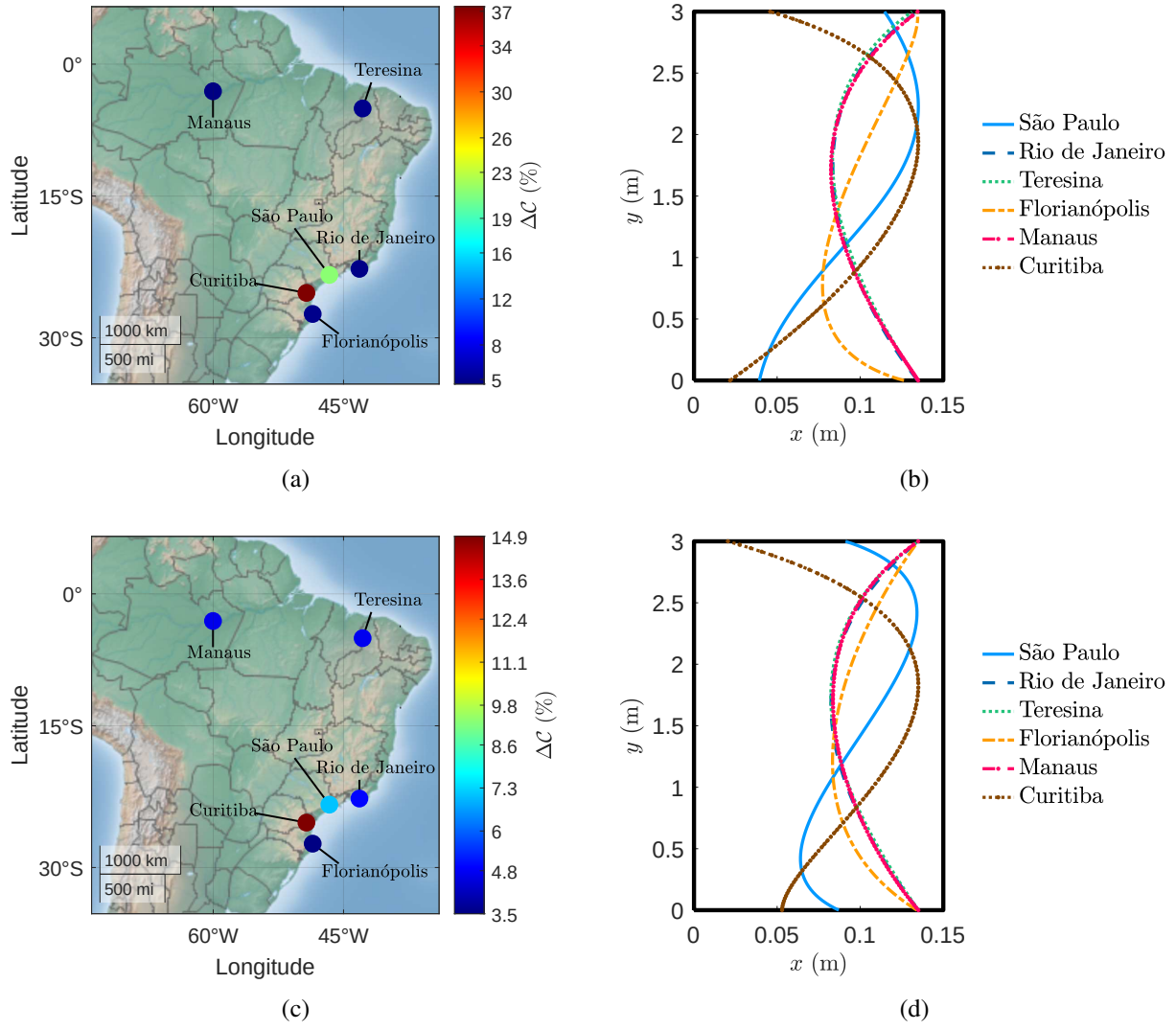


Fig. 7 Illustration of energy efficiency improvements according to Brazilian geography for north-facing (a) and west-facing (c) orientations and their respective optimal interfaces (b, d) in IWI configuration

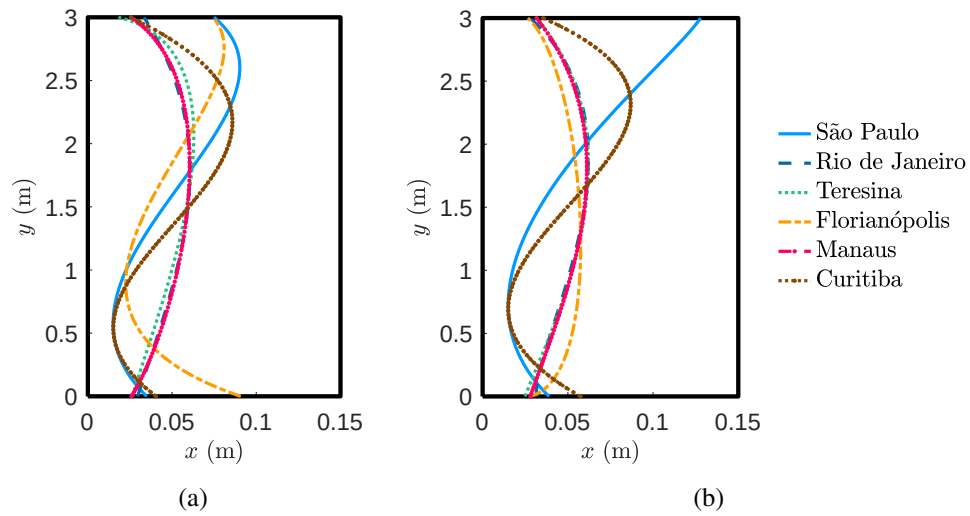


Fig. 8 Illustration of the optimal interfaces in EWI configuration for north (a) and west (b) facing orientations.

Table 2 *Improvement of the optimal interfaces with IWI and EWI configurations according to climatic conditions and façade orientation.*

wall insulation configuration - IWI						
<i>orientation</i>	<i>Curitiba</i>	<i>Florianópolis</i>	<i>Manaus</i>	<i>Rio de Janeiro</i>	<i>São Paulo</i>	<i>Teresina</i>
north	37.37%	4.90%	4.62%	4.80%	21.30%	4.89%
west	14.87%	3.48%	4.66%	4.95%	7.10%	4.74%
wall insulation configuration - EWI						
<i>orientation</i>	<i>Curitiba</i>	<i>Florianópolis</i>	<i>Manaus</i>	<i>Rio de Janeiro</i>	<i>São Paulo</i>	<i>Teresina</i>
north	7.61%	5.12%	1.54%	1.69%	16.26%	2.16%
west	5.39%	2.97%	1.62%	1.80%	16.60%	1.99%

5. CONCLUSIONS

This study demonstrates that optimizing the geometry of insulation interfaces in building walls can significantly improve thermal performance by adapting to realistic boundary conditions. By accounting for transient effects and spatial variations in wind and solar radiation, the proposed numerical framework identifies non-planar interface geometries that outperform conventional flat designs.

The key finding is that internal wall insulation (IWI) generally leads to larger energy improvements compared to external wall insulation (EWI). Climatic conditions also play a decisive role: very hot climates with intense solar exposure tend to yield smaller improvements than more temperate climates with moderate sunlight. This highlights the need for region specific optimization. Moreover, the proposed interface design proves beneficial for multiple façade orientations, as observed for the two orientations evaluated. Overall, the proposed approach is capable of delivering substantial gains, reaching nearly 40% annual improvement in the best cases considered in this study.

Although this work focuses on interface optimization using BÉZIER curves defined by $N_p = 4$ particles, additional control points and alternative interface definition could be employed to describe more complex geometries. Future research should investigate these extended design parameters to broaden the accessible design space. Moreover, the optimization criteria themselves could evolve toward multi-objective formulations, and the overall geometry of the wall could also be reconsidered to further enhance the thermal efficiency of building envelopes. Subsequently, the internal and external boundary conditions can be refined to include long-wavelength radiation.

Finally, although the proposed approach provides valuable information at the wall level, it remains subject to inherent limitations. The optimal configurations obtained should therefore be interpreted with caution, as it is not straightforward to extrapolate them directly to the performance of the entire building. Future research could therefore explore the integration of this approach at the building level.

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NOMENCLATURE

c	volumetric heat capacity	(J.m ³ .K ⁻¹)	N_y	y space nodes number	(-)
$\mathcal{C}$	cost function	(J)	$f_{\mathcal{B}}$	interface functions	(m)
h	shadow height	(m)	Γ	boundary	(-)
h	heat transfer coefficient	(W.m ⁻² .K ⁻¹)	Δt	time step	(s)
H	building facade height	(m)	Δx	x space mesh	(m)
L	wall length	(m)	Δy	y space mesh	(m)
N_p	number of parameters	-	ΔX	canyon street width	(m)
$N_{\bullet}$	number of particles	(-)	ΔY	canyon street height	(m)
$\mathbf{p}$	interface parameters	(-)	λ	heat conductivity	(W.m ⁻¹ .K ⁻¹)
q	thermal flux	(W.m ⁻²)	Ω_p	interface parameters domain	(-)
t_f	final time	(s)	Ω	space domain	(-)
N_t	number of time instants	(-)	N_i	number of individuals	(-)
T	temperature	(K)	N_g	number of generations	(-)
v_{∞}	wind velocity	(m.s ⁻¹)	c_r	crossover rate	(-)
x	horizontal space coordinate	(m)	m_r	mutation rate	(-)
y	vertical space coordinate	(m)	IWI	Internal Wall Insulation	
N_x	x space nodes number	(-)	EWI	External Wall Insulation	

REFERENCES

- [1] IEA, P., “World energy outlook 2022”, *Paris, France: International Energy Agency (IEA)* (2022).
- [2] Berger, J., Gasparin, S., Mazuroski, W. and Mendes, N., “An efficient two-dimensional heat transfer model for building envelopes”, *Numerical Heat Transfer, Part A: Applications*, 79(3), pp. 163–194, doi:10.1080/10407782.2020.1836936, Available at <https://doi.org/10.1080/10407782.2020.1836936> (2020).
- [3] Vantighem, G., De Corte, W., Steeman, M. and Boel, V., “Density-based topology optimization for 3D-printable building structures”, *Structural and Multidisciplinary Optimization*, 60(6), pp. 2391–2403, doi:10.1007/s00158-019-02330-7, Available at <https://doi.org/10.1007/s00158-019-02330-7> (2019).
- [4] Sousa, L. C., Castro, C. F., António, C. C. and Sousa, H., “Topology optimisation of masonry units from the thermal point of view using a genetic algorithm”, *Construction and Building Materials*, 25(5), pp. 2254–2262, doi:10.1016/j.conbuildmat.2010.11.010, Available at <https://www.sciencedirect.com/science/article/pii/S0950061810005222> (2011).
- [5] Vantighem, G., Boel, V., Steeman, M. and De Corte, W., “Multi-material topology optimization involving simultaneous structural and thermal analyses”, *Structural and Multidisciplinary Optimization*, 59(3), pp. 731–743, doi:10.1007/s00158-018-2095-z, Available at <https://doi.org/10.1007/s00158-018-2095-z> (2019).
- [6] Karashbayeva, Z., Gasparin, S. and Berger, J., “Optimisation of the insulation domain for heat transfer in building wall”, *11th INTERNATIONAL CONFERENCE ON INVERSE PROBLEMS IN ENGINEERING: THEORY AND PRACTICE (ICIPE 2024)*, Buzios, Brazil, Available at <https://hal.science/hal-04660099> (2024).
- [7] Alpar, S., Berger, J. and Belarbi, R., “Shape optimisation of the building facade energy efficiency in transient regime”, *11th INTERNATIONAL CONFERENCE ON INVERSE PROBLEMS IN ENGINEERING: THEORY AND PRACTICE*, Buzios - Rio de Janeiro, Brazil, Available at <https://hal.science/hal-04659258> (2024).
- [8] Alpar, S., Berger, J., Mazuroski, W. and Belarbi, R., “Shape optimization of the energy efficiency of building retrofitted facade”, *Solar Energy*, 271, p. 112437, doi:10.1016/j.solener.2024.112437, Available at <https://www.sciencedirect.com/science/article/pii/S0038092X24001312> (2024).
- [9] Deb, K., Pratap, A., Agarwal, S. and Meyarivan, T., “A fast and elitist multiobjective genetic algorithm: NSGA-II”, *IEEE Transactions on Evolutionary Computation*, 6(2), pp. 182–197, doi:10.1109/4235.996017, Available at <http://ieeexplore.ieee.org/document/996017/> (2002).
- [10] William H. Mcadams, *Heat Transmission*, Available at <http://archive.org/details/cftri10979heattransmission0000will> (1954).
- [11] Rocha, A. P. d. A., Oliveira, R. C. L. F. and Mendes, N., “Experimental validation and comparison of direct solar shading calculations within building energy simulation tools: Polygon clipping and pixel counting techniques”, *Solar Energy*, 158, pp. 462–473, doi:10.1016/j.solener.2017.10.011, Available at <https://www.sciencedirect.com/science/article/pii/S0038092X17308769> (2017).
- [12] dos Santos, G. H., Mendes, N. and Philippi, P. C., “A building corner model for hygrothermal performance and mould growth risk analyses”, *International Journal of Heat and Mass Transfer*, 52(21), pp. 4862–4872, doi:10.1016/j.ijheatmasstransfer.2009.05.026, Available at <https://www.sciencedirect.com/science/article/pii/S0017931009003561> (2009).
- [13] Dumontaud, W., Berger, J. and Kadoch, B., “A Novel 2d Diffusion Model for Composite Domain with Non-Regularly Shaped Interface”, doi:10.2139/ssrn.5369908, Available at <https://papers.ssrn.com/abstract=5369908> (2025).

- [14] Fedkiw, R. P., Aslam, T., Merriman, B. and Osher, S., “A Non-oscillatory Eulerian Approach to Interfaces in Multimaterial Flows (the Ghost Fluid Method)”, *Journal of Computational Physics*, 152(2), pp. 457–492, doi:10.1006/jcph.1999.6236, Available at <https://www.sciencedirect.com/science/article/pii/S0021999199962368> (1999).
- [15] Strub, F., Castaing-Lasvignottes, J., Strub, M., Pons, M. and Monchoux, F., “Second law analysis of periodic heat conduction through a wall”, *International Journal of Thermal Sciences*, 44(12), pp. 1154–1160, doi:10.1016/j.ijthermalsci.2005.09.004, Available at <https://linkinghub.elsevier.com/retrieve/pii/S1290072905002127> (2005).
- [16] “Meteonorm”, Available at <https://meteonorm.com/en/>.